

1. Simulate a DNS resolution failure on your local machine by editing the hosts file to point `www.spotify.com` to a wrong IP address, then try to ping the domain and document the error message you receive.

Simulate a DNS Resolution Failure Using the Windows hosts File

Note: This changes only your local machine's name resolution. It does **not** change Spotify's DNS records.

1. Open the Windows Hosts File

The hosts file is located at:

`C:\Windows\System32\drivers\etc\hosts`

1. Open **Notepad as Administrator**.
2. Select **File** → **Open**.
3. Navigate to:

`C:\Windows\System32\drivers\etc`

4. Change the file filter from **Text Documents (*.txt)** to **All Files**.
5. Open the file named **hosts**.

2. Add a Wrong IP Address

Add this line at the bottom:

`203.0.113.50 www.spotify.com`

203.0.113.50 is an example/documentation IP address and is intentionally not the real Spotify server address.

Save the file.

3. Clear the DNS Cache

Open **Command Prompt as Administrator** and run:

`ipconfig /flushdns`

You should see:

Windows IP Configuration

Successfully flushed the DNS Resolver Cache.

4. Ping Spotify

Run:

```
ping www.spotify.com
```

Because the hosts file overrides normal DNS resolution, Windows should resolve `www.spotify.com` to the incorrect address.

You may see something similar to:

```
Pinging www.spotify.com [203.0.113.50] with 32 bytes of data:
```

```
Request timed out.
```

```
Request timed out.
```

```
Request timed out.
```

```
Request timed out.
```

```
Ping statistics for 203.0.113.50:
```

```
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)
```

Important: The exact error can vary. If the wrong IP is unreachable, you may get "**Request timed out.**" If the address responds for some reason, you may instead receive replies. The key evidence is that the ping shows:

```
www.spotify.com [203.0.113.50]
```

rather than Spotify's normal resolved address.

5. Verify the Incorrect Resolution

Run:

```
nslookup www.spotify.com
```

nslookup normally queries DNS directly and may show a different result from the hosts-file-based resolution used by ping.

For a clearer Windows test, use:

```
ping www.spotify.com
```

and observe the IP displayed beside the hostname.

6. Restore the Hosts File

After completing the experiment, **remove the line**:

```
203.0.113.50 www.spotify.com
```

Save the hosts file and run:

```
ipconfig /flushdns
```

Then test again:

```
ping www.spotify.com
```

Lab Result

Test	Result
Hosts entry	203.0.113.50 www.spotify.com
DNS cache	Flushed
Ping result	Resolves to the incorrect IP
Typical error	Request timed out.
Cause	Local hosts file overrides normal DNS resolution
After removing entry	Normal DNS resolution is restored

2. Configure a DHCP server in a network simulator (such as Cisco Packet Tracer or GNS3) and intentionally set an incorrect default gateway; connect a client device and verify the assigned gateway, then describe how you would identify and fix this issue.

DHCP Server with an Incorrect Default Gateway — Cisco Packet Tracer

1. Create the Network

Use these devices in **Cisco Packet Tracer**:

- 1 × Server
- 1 × Switch
- 1 × PC
- Copper Straight-Through cables

Connect:

PC0 ——— Switch0 ——— Server0

2. Configure the Server

Click **Server0** → **Desktop** → **IP Configuration**.

Set:

Setting	Value
IP Address	192.168.10.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1

Now go to **Services** → **DHCP**.

Turn DHCP **On** and create a pool:

DHCP Setting	Value
Pool Name	LAN_POOL
Default Gateway	192.168.10.99 ✕
DNS Server	8.8.8.8
Start IP Address	192.168.10.10
Subnet Mask	255.255.255.0
Maximum Users	50

Here, 192.168.10.99 is intentionally configured as the **wrong gateway**.

Click **Add/Save**.

3. Configure the Client

Click **PC0** → **Desktop** → **IP Configuration**.

Select:

DHCP

The PC should receive something similar to:

IP Address: 192.168.10.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.10.99

DNS Server: 8.8.8.8

The important observation is:

PC0 received 192.168.10.99 as its default gateway, even though the correct gateway is 192.168.10.1.

4. Verify Using Command Prompt

Open:

PC0 → Desktop → Command Prompt

Run:

ipconfig

You should see:

IP Address.....: 192.168.10.10

Subnet Mask.....: 255.255.255.0

Default Gateway.....: 192.168.10.99

You can also test the gateway:

ping 192.168.10.99

The ping will fail if no device is actually using 192.168.10.99.

Test the correct gateway:

ping 192.168.10.1

If a router/interface exists at 192.168.10.1, it should respond.

5. How to Identify the Problem

Check the client configuration first:

ipconfig

Notice that the **Default Gateway** is 192.168.10.99.

Then compare it with the actual router/interface address:

Correct gateway: 192.168.10.1

Assigned gateway: 192.168.10.99

This indicates that the DHCP server is providing an incorrect gateway.

6. Fix the DHCP Configuration

Go back to:

Server0 → Services → DHCP → LAN_POOL

Change:

Default Gateway:

192.168.10.99

to:

Default Gateway:

192.168.10.1

Save the configuration.

7. Renew the Client's DHCP Address

On PC0, open Command Prompt:

```
ipconfig /release
```

```
ipconfig /renew
```

Then check:

```
ipconfig
```

The result should now be:

IP Address.....: 192.168.10.10

Subnet Mask.....: 255.255.255.0

Default Gateway.....: 192.168.10.1

Final Verification

Run:

```
ping 192.168.10.1
```

Then, if the simulated network has Internet/WAN connectivity, test an external destination.

Summary

Item	Before Fix ✕	After Fix ✓
Client IP	192.168.10.10	192.168.10.10
Subnet Mask	255.255.255.0	255.255.255.0
Default Gateway	192.168.10.99	192.168.10.1
Problem	Wrong DHCP gateway	Correct gateway

Result	External communication fails	Routing works
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3. Given a scenario where devices on your simulated network are unable to access websites by name but can by IP address, use the nslookup or dig tool to diagnose the DNS problem and write down the troubleshooting steps and solution.

Scenario A: Public DNS worked (Step 2), but local failed

* **The Fix:** Your local DNS server is down or broken. Restart the DNS service (like BIND or Windows DNS) on your simulated server.

* **The Workaround:** If you can't fix the server right away, update your network's DHCP settings to hand out a reliable public DNS server (like `8.8.8.8`) to all client devices instead of your local one.

Scenario B: Everything failed, even public DNS (Step 2)

* **The Fix:** Your network's router or firewall is dropping DNS traffic. Check the firewall rules on your simulated router and ensure that outbound traffic on **Port 53 (UDP and TCP)** is allowed to pass through to the internet.

Scenario C: The client had no DNS server listed (Step 3)

* **The Fix:** Your DHCP server is misconfigured and isn't assigning a DNS server to devices when they join the network. Check your DHCP scope options and ensure the "DNS Server" option (Option 6) is filled out with a valid IP address.

4. A device in your simulated network is not receiving an IP address from the DHCP server and shows an APIPA address (169.x.x.x). List the possible causes for this and document the step-by-step process you would use to resolve it.

DHCP Troubleshooting: Device Has a 169.x.x.x APIPA Address

Scenario

A client device is supposed to receive an IP address from a DHCP server, but instead it receives an **APIPA address**, such as:

IP Address: 169.254.10.25

Subnet Mask: 255.255.0.0

An address in the **169.254.0.0/16** range usually means the device could not obtain a valid IPv4 address from DHCP.

Possible Causes

Cause	Explanation
1. DHCP server is turned off	The DHCP service may not be running.
2. DHCP pool is exhausted	No available addresses remain in the DHCP pool.
3. Incorrect DHCP configuration	Wrong network, subnet mask, gateway, or address range may be configured.
4. Cable/Wi-Fi connection problem	The client cannot communicate with the network.
5. Wrong VLAN	The client may be connected to a VLAN that cannot reach the DHCP server.
6. DHCP relay missing	If the DHCP server is on another subnet, a router may need DHCP relay (ip helper-address).
7. DHCP server unreachable	Routing between the client and DHCP server may be incorrect.
8. Duplicate/conflicting configuration	Another DHCP server or incorrect static configuration can interfere with DHCP.
9. Network interface problem	The client's network adapter may be disabled or incorrectly configured.

Step-by-Step Troubleshooting

Step 1: Check the Client IP Address

On the simulated PC, open:

Desktop → Command Prompt

Run:

ipconfig

If you see:

IP Address: 169.254.x.x

the client did not receive a DHCP lease.

Step 2: Check the Physical Connection

In Cisco Packet Tracer:

- Make sure the PC is connected to the switch.
- Check that the cable is correct.
- Check that link indicators are green.
- Make sure the switch port is enabled.

For a Cisco switch, you can check:

Switch# show interfaces status

Look for the port connected to the PC.

Step 3: Check the Client's DHCP Setting

On the PC:

Desktop → IP Configuration

Select:

DHCP

Do not use a manually configured IP address.

Then try:

ipconfig /renew

In Packet Tracer, simply selecting **DHCP** again can trigger another DHCP request.

Step 4: Check the DHCP Server

Go to:

Server → Services → DHCP

Make sure DHCP is:

DHCP: ON

Check the configured pool.

Example:

Setting	Correct Value
Network	192.168.10.0
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1
Start IP	192.168.10.10
Maximum Users	50

Make sure the pool still has available addresses.

Step 5: Check the DHCP Server's Own IP

The DHCP server should have a **static IP**.

Example:

IP Address: 192.168.10.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.10.1

Do not configure the DHCP server itself to obtain its address from the same DHCP pool.

Step 6: Test Basic Connectivity

From the client, test the gateway:

ping 192.168.10.1

If the client still has an APIPA address, this may fail because it is not properly connected to the intended subnet.

If the server and client are on the same LAN, verify that they can communicate at Layer 2.

Step 7: Check VLAN Configuration

If VLANs are being used, verify that the PC's switch port is in the correct VLAN.

For example:

Switch# show vlan brief

Make sure the PC's port belongs to the correct VLAN.

Example:

VLAN 10

PC0 → Fa0/1

DHCP Server → VLAN 10

A PC placed in the wrong VLAN may not reach the DHCP server.

Step 8: If DHCP Server Is on Another Network, Check DHCP Relay

If the client and DHCP server are on different subnets, configure DHCP relay on the router interface.

Example:

```
Router(config)# interface gigabitEthernet0/0
```

```
Router(config-if)# ip helper-address 192.168.20.2
```

```
Router(config-if)# no shutdown
```

Here, 192.168.20.2 is the DHCP server's IP address.

The ip helper-address forwards DHCP requests from the client network to the DHCP server.

Step 9: Check for an Incorrect DHCP Pool

For example, suppose the client is on:

192.168.10.0/24

but the DHCP server is configured to distribute:

192.168.20.0/24

This is incorrect.

The DHCP pool must match the client's network.

Step 10: Request an IP Address Again

After correcting the problem:

```
ipconfig /release
```

```
ipconfig /renew
```

Then:

```
ipconfig
```

The client should now receive something like:

IP Address: 192.168.10.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.10.1

The **169.254.x.x** address should be gone.

Final Verification

Test the gateway:

ping 192.168.10.1

Then test another device on the network:

ping 192.168.10.2

If the network has DNS/Internet connectivity, test:

ping google.com

5. Create a table documenting at least three common DNS issues and three common DHCP issues you encountered or researched, including symptoms, possible causes, and recommended solutions for each.

Type	Issue	Symptoms	Possible Causes	Recommended Solution
DNS	DNS server unreachable	Websites do not open by domain name; nslookup shows timeout	Incorrect DNS IP, DNS server down, network connectivity problem	Check DNS address with ipconfig /all, ping the DNS server, and configure a working DNS server
DNS	Incorrect DNS record	Domain resolves to the wrong IP address	Incorrect A/AAAA record or outdated DNS record	Check with nslookup/dig and correct the DNS record on the DNS server
DNS	DNS cache problem	Old or incorrect IP address continues to appear	Stale local DNS cache	Flush the cache using ipconfig /flushdns, then test again

DHCP	Client receives APIPA address (169.254.x.x)	Device has no normal network IP and cannot communicate properly	DHCP server unavailable, network connection failure, DHCP request blocked	Check network connection, verify DHCP service is running, and renew the DHCP lease
DHCP	DHCP pool exhausted	New devices cannot obtain an IP address	All addresses in the DHCP pool are already allocated	Expand the DHCP address pool or remove unused leases
DHCP	Incorrect DHCP configuration	Client receives an IP but cannot reach other networks/Internet	Wrong subnet mask, default gateway, or DNS server in the DHCP pool	Correct the DHCP scope options and renew the client's lease